Lab 7: Fuzzing for Input Validation Bugs

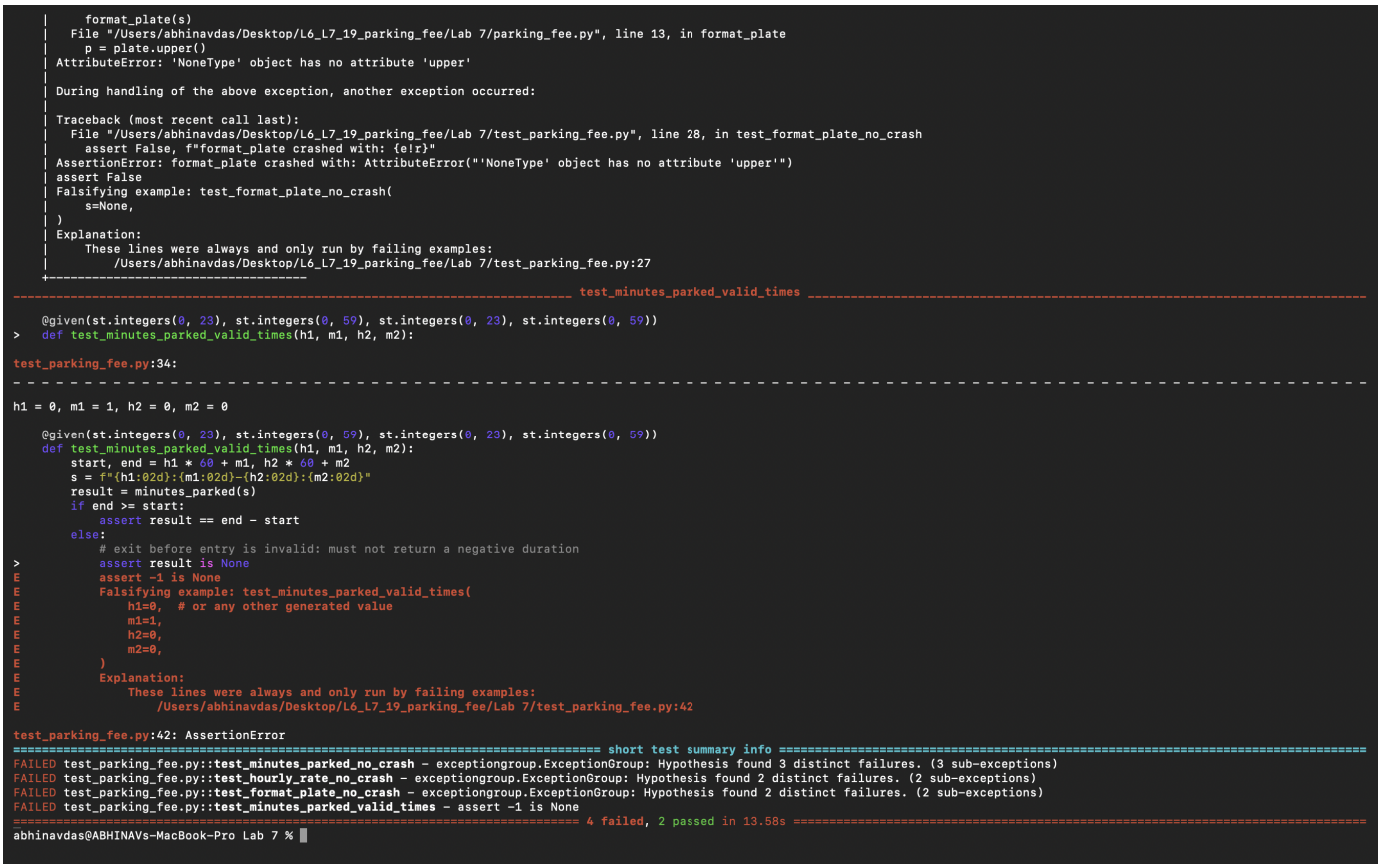
PES2UG24CS022 | Repository: SETAPESU26/L6_L7_19_parking_fee | Module: parking_fee.py

Step 2: Running the buggy application (python3 parking_fee.py)

```
==== Running minutes_parked ====
Enter entry and exit time (e.g. 09:30-11:45): hello
Traceback (most recent call last):
  File "parking_fee.py", line 33, in <module>
    main()
  File "parking_fee.py", line 19, in main
    result = minutes_parked(user_input)
  File "parking_fee.py", line 4, in minutes_parked
    entry, exit_time = times.split('-')
ValueError: not enough values to unpack (expected 2, got 1)
abhinavdas@ABHINAVs-MacBook-Pro Lab 7 %
```

Input 'hello' crashes minutes_parked() with ValueError: not enough values to unpack.

Step 4: Running the Hypothesis fuzz tests on the buggy code (pytest test_parking_fee.py)



```
    |     format_plate(s)
    |   File "/Users/abhinavdas/Desktop/L6_L7_19_parking_fee/Lab 7/parking_fee.py", line 13, in format_plate
    |     p = plate.upper()
    | AttributeError: 'NoneType' object has no attribute 'upper'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/Users/abhinavdas/Desktop/L6_L7_19_parking_fee/Lab 7/test_parking_fee.py", line 28, in test_format_plate_no_crash
    |     assert False, f"format_plate crashed with: {e!r}"
    | AssertionError: format_plate crashed with: AttributeError("'NoneType' object has no attribute 'upper'")
    | assert False
    | Falsifying example: test_format_plate_no_crash(
    |     s=None,
    | )
    | Explanation:
    |     These lines were always and only run by failing examples:
    |         /Users/abhinavdas/Desktop/L6_L7_19_parking_fee/Lab 7/test_parking_fee.py:27
    +------------------------------------
_____________________________________________________________________ test_minutes_parked_valid_times _____________________________________________________________________

    @given(st.integers(0, 23), st.integers(0, 59), st.integers(0, 23), st.integers(0, 59))
>   def test_minutes_parked_valid_times(h1, m1, h2, m2):

test_parking_fee.py:34:
- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -

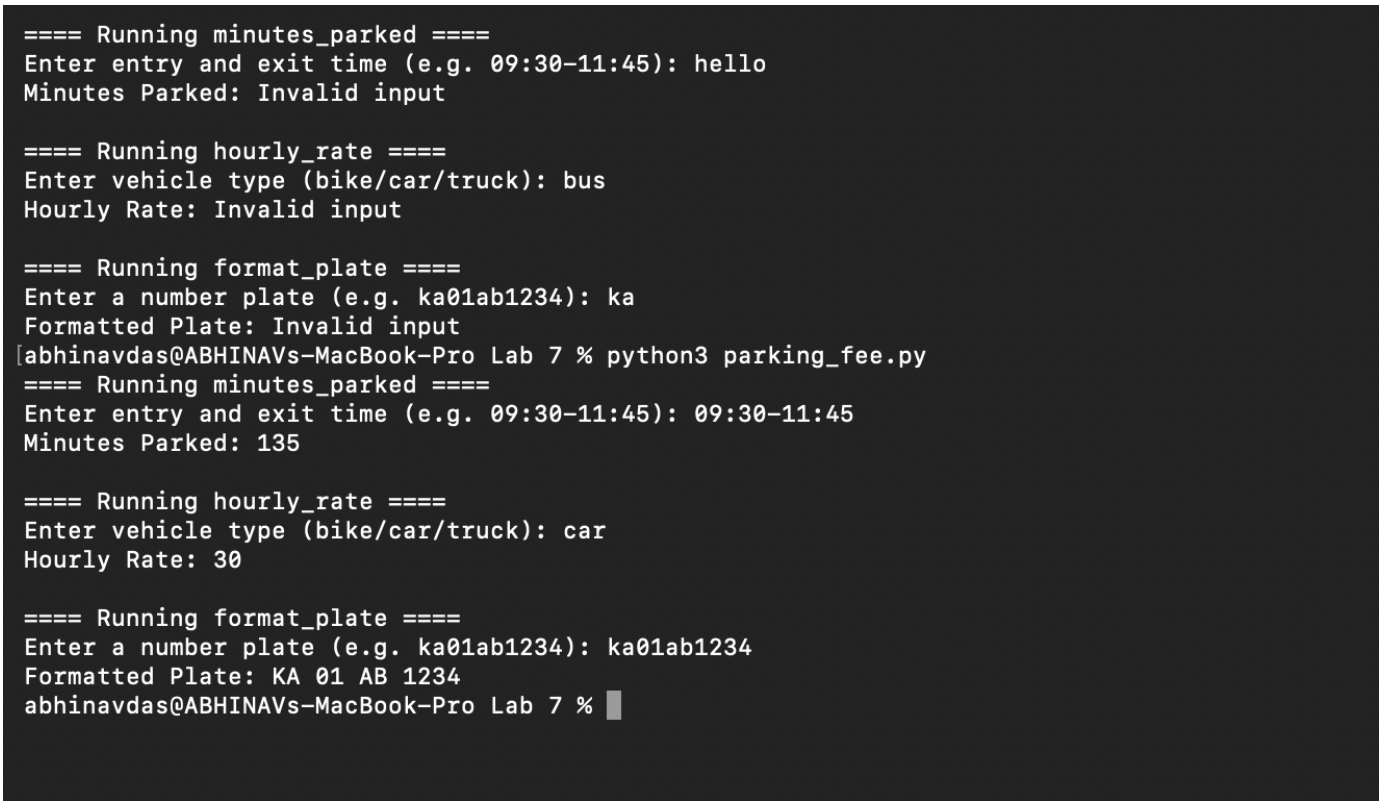
h1 = 0, m1 = 1, h2 = 0, m2 = 0

    @given(st.integers(0, 23), st.integers(0, 59), st.integers(0, 23), st.integers(0, 59))
    def test_minutes_parked_valid_times(h1, m1, h2, m2):
        start, end = h1 * 60 + m1, h2 * 60 + m2
        s = f"{h1:02d}:{m1:02d}-{h2:02d}:{m2:02d}"
        result = minutes_parked(s)
        if end >= start:
            assert result == end - start
        else:
            # exit before entry is invalid: must not return a negative duration
>           assert result is None
E           assert -1 is None
E           Falsifying example: test_minutes_parked_valid_times(
E               h1=0,  # or any other generated value
E               m1=1,
E               h2=0,
E               m2=0,
E           )
E           Explanation:
E               These lines were always and only run by failing examples:
E                   /Users/abhinavdas/Desktop/L6_L7_19_parking_fee/Lab 7/test_parking_fee.py:42

test_parking_fee.py:42: AssertionError
=========================================================================== short test summary info ============================================================================
FAILED test_parking_fee.py::test_minutes_parked_no_crash - exceptiongroup.ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
FAILED test_parking_fee.py::test_hourly_rate_no_crash - exceptiongroup.ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
FAILED test_parking_fee.py::test_format_plate_no_crash - exceptiongroup.ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
FAILED test_parking_fee.py::test_minutes_parked_valid_times - assert -1 is None
========================================================================= 4 failed, 2 passed in 13.58s =========================================================================
abhinavdas@ABHINAVs-MacBook-Pro Lab 7 %
```

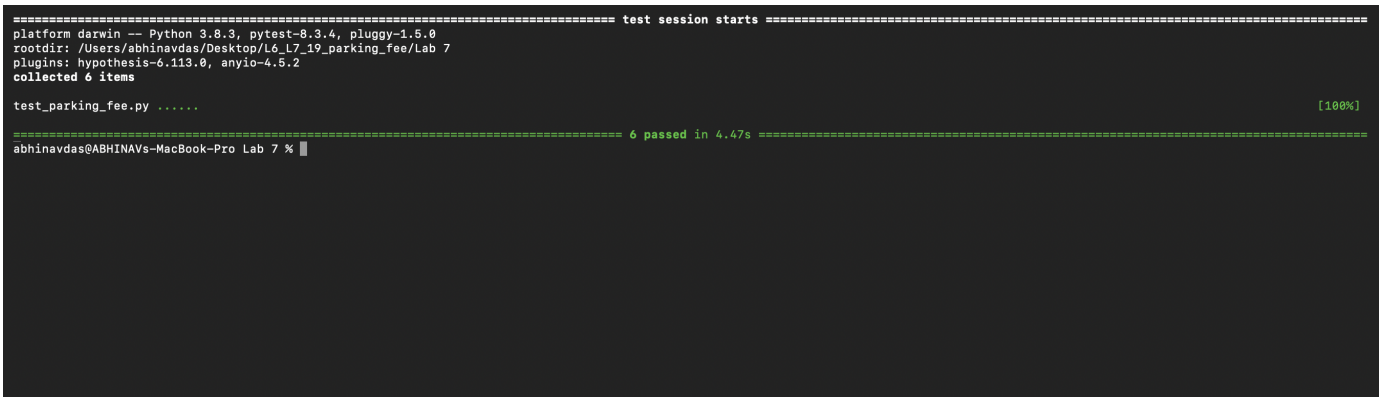
4 failed, 2 passed. Hypothesis found crashes for None, empty and malformed strings in all three functions (ValueError, AttributeError, KeyError, IndexError) and a negative duration when exit time is before entry time (e.g. 00:01-00:00 returns -1).

Step 5: Running the fixed application (python3 parking_fee.py)



Invalid inputs ('hello', 'bus', 'ka') now print 'Invalid input' instead of crashing; valid inputs still give 135 minutes, rate 30 and plate KA 01 AB 1234.

Step 6: Re-running the fuzz tests on the fixed code (pytest test_parking_fee.py)



All 6 tests pass.